

New Wheels Project

Introduction to SQL

Problem Statement

Business Context

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Questions

Question 1:

Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
select b.state,count(a.customer_id) as count_of_customers_per_state from order_t a
inner join customer_t b
on a.customer_id=b.customer_id
group by b.state order by count_of_customers_per_state desc
```

Output:

```
1 • select b.state,count(a.customer_id) as count_of_customers_per_state from order_t a
2   inner join customer_t b
3   on a.customer_id=b.customer_id
4   group by b.state order by count_of_customers_per_state desc
```

Result Grid	
Filter Rows:	Export: Wrap Cell Content:
state	count_of_customers_per_state
Texas	98
California	97
Florida	86
New York	69
District of Columbia	35
Ohio	33
...	...

Observations and Insights:

1. Texas and California dominate. With 98 and 97 orders respectively, these two states are virtually tied at the top and together account for a massive share of the customer base. New Wheels' business is heavily concentrated in large, populous Sun Belt states.

2. Florida rounds out the Big Three. At 86 orders, Florida forms a clear top tier with Texas and California. These three states alone contribute the majority of all orders visible in the result.

3. Significant dropoff after the top 3. New York (69), Wyoming (1), and Maine (1) show a notable stepdown in volume, suggesting the brand has a strong West/South bias rather than uniform national reach.

Question 2:

Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
WITH Final as (  
  select t.*, row_number()over(order by t.vehicle_maker_count desc) as rownum  
  from  
    (select b.vehicle_maker, count(*) as vehicle_maker_count from order_t a  
     inner join product_t b  
     on a.product_id=b.product_id  
     group by b.vehicle_maker) t  
 )  
SELECT * FROM FINAL where rownum<=5
```

Output:

```
1 • WITH Final as (  
2   select t.*, row_number()over(order by t.vehicle_maker_count desc) as rownum  
3   from  
4   (select b.vehicle_maker,count(*) as vehicle_maker_count from order_t a  
5    inner join product_t b  
6    on a.product_id=b.product_id  
7    group by b.vehicle_maker) t  
8  )  
9  SELECT * FROM FINAL where rownum<=5
```

vehicle_maker	vehicle_maker_count	rownum
Chevrolet	83	1
Ford	63	2
Toyota	52	3
Dodge	50	4
Pontiac	50	5

Observations and Insights:

1. Chevrolet is the clear market leader. With 83 orders, Chevrolet leads by a significant margin 20 orders ahead of Ford (63). This suggests strong customer loyalty or a broader product range that appeals to New Wheels customer base.
2. All top 5 are American brands. Chevrolet, Ford, Dodge, and Pontiac are all GM or Ford Motor Company brands, with Toyota being the only nonAmerican entry. This points to a predominantly American vehicle preference among customers.
3. There's a visible two tier pattern. Chevrolet and Ford form a dominant top tier (63 to 83 orders), while Toyota, Dodge, and Pontiac cluster tightly in a second tier (50 to 52), suggesting a natural segmentation in customer preference. Dodge and Pontiac are tied at 50.

Question 3:

Which is the most preferred vehicle maker in each state?

Solution Query:

```
WITH Final as (  
  select t.*, rank()over(partition by t.state order by t.customer_count desc) as rownum  
  from  
    (select c.state,b.vehicle_maker,count(a.CUSTOMER_ID) as customer_count from order_t a  
     inner join product_t b  
       on a.product_id=b.product_id  
     inner join customer_t c  
       on a.customer_id = c.customer_id  
     group by c.state,b.vehicle_maker) t  
)  
SELECT state,vehicle_maker FROM FINAL where rownum=1
```

Output:

```
1 • WITH Final as (  
2   select t.*, rank()over(partition by t.state order by t.customer_count desc) as rownum  
3   from  
4   (select c.state,b.vehicle_maker,count(a.CUSTOMER_ID) as customer_count from order_t a  
5    inner join product_t b  
6    on a.product_id=b.product_id  
7    inner join customer_t c  
8    on a.customer_id = c.customer_id  
9    group by c.state,b.vehicle_maker) t  
10  )  
11  SELECT state,vehicle_maker FROM FINAL where rownum=1  
12
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	state	vehide_maker
•	Alabama	Dodge
	Alaska	Chevrolet
	Arizona	Pontiac
	Arizona	Cadillac
	Arkansas	Suzuki
	Arkansas	Chevrolet
	Arkansas	Pontiac

Observations and Insights:

1. Chevrolet dominates across the nation. It is the 1 preferred maker in the largest number of states covering a wide spread from Texas and California to Montana and Wisconsin. Its national reach is unmatched in this dataset.
2. Ford is a strong regional runner up. Ford leads in states like Michigan (its home state), New York, Illinois, Washington, and Kentucky concentrated in the Midwest and coastal regions.
3. **Niche makers appear in smaller states.** Suzuki appearing in Arkansas and Cadillac in Arizona suggests low volume states are more sensitive to just a few orders tipping the balance even 1 and 2 orders can crown a maker in a lightly ordered state.

Question 4:

Find the overall average rating given by the customers. What is the average rating in each quarter?

Solution Query:

```
with feedback_rating as (  
  select customer_id, quarter_number, case when customer_feedback='Very Bad' then 1  
    when customer_feedback='Bad' then 2  
    when customer_feedback='Okay' then 3  
    when customer_feedback='Good' then 4  
    when customer_feedback='Very Good' then 5 else 0 End as rating  
  from order_t  
)  
SELECT avg(rating), quarter_number from feedback_rating  
group by quarter_number
```

Output:

```
1 with feedback_rating as (  
2   select customer_id,quarter_number,case when customer_feedback='Very Bad' then 1  
3     when customer_feedback='Bad' then 2  
4     when customer_feedback='Okay' then 3  
5     when customer_feedback='Good' then 4  
6     when customer_feedback='Very Good' then 5 else 0 End as rating  
7   from order_t  
8 )  
9 SELECT avg(rating),quarter_number from feedback_rating  
10 group by quarter_number  
11
```

Result Grid Filter Rows: Export: Wrap Cell Content:		
	avg(rating)	quarter_number
▶	2.3970	4
	3.5548	1
	2.9563	3
	3.3550	2

Observations & Insights

1. Clear and consistent decline across all quarters. Ratings drop steadily from Q1 (3.55) ,Q2 (3.36) , Q3 (2.96) ,Q4 (2.40). This is not a random fluctuation it's a sustained downward trend throughout the year, which is a serious signal.
2. Overall average is 3.08 / 5 just barely "Okay". The overall annual rating sits right at the midpoint of the scale, meaning customer sentiment is neutral at best. There is significant room for improvement.
3. Q4 is critically low at 2.40 approaching "Bad". A rating under 2.5 means the majority of Q4 customers leaned towards "Bad" or "Very Bad" feedback. This quarter demands urgent attention from operations and customer service teams.
4. Q1 is the only quarter above "Okay" (3.0). Only Q1 exceeds the neutral threshold, suggesting the business starts the year strong but struggles to maintain service quality as the year

progresses possibly due to seasonal demand pressure, inventory issues, or shipping delays in later quarters.

Question 5:

Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
WITH feedback_rating AS ( SELECT customer_id, quarter_number, CASE WHEN  
customer_feedback = 'Very Bad' THEN 1 WHEN customer_feedback = 'Bad' THEN 2 WHEN  
customer_feedback = 'Okay' THEN 3 WHEN customer_feedback = 'Good' THEN 4 WHEN  
customer_feedback = 'Very Good' THEN 5 ELSE 0 END AS rating FROM order_t )
```

```
SELECT quarter_number,
```

```
COUNT(*) AS total_feedback,
```

```
SUM(CASE WHEN rating = 1 THEN 1 ELSE 0 END) * 100.0 / COUNT(*) AS very_bad_pct,
```

```
SUM(CASE WHEN rating = 2 THEN 1 ELSE 0 END) * 100.0 / COUNT(*) AS bad_pct,
```

```
SUM(CASE WHEN rating = 3 THEN 1 ELSE 0 END) * 100.0 / COUNT(*) AS okay_pct,
```

```
SUM(CASE WHEN rating = 4 THEN 1 ELSE 0 END) * 100.0 / COUNT(*) AS good_pct,
```

```
SUM(CASE WHEN rating = 5 THEN 1 ELSE 0 END) * 100.0 / COUNT(*) AS very_good_pct
```

```
FROM feedback_rating GROUP BY quarter_number ORDER BY quarter_number;
```

Output:

```
2
3  SELECT quarter_number,
4
5  COUNT(*) AS total_feedback,
6
7  SUM(CASE WHEN rating = 1 THEN 1 ELSE 0 END) * 100.0 / COUNT(*) AS very_bad_pct,
8  SUM(CASE WHEN rating = 2 THEN 1 ELSE 0 END) * 100.0 / COUNT(*) AS bad_pct,
9  SUM(CASE WHEN rating = 3 THEN 1 ELSE 0 END) * 100.0 / COUNT(*) AS okay_pct,
10 SUM(CASE WHEN rating = 4 THEN 1 ELSE 0 END) * 100.0 / COUNT(*) AS good_pct,
11 SUM(CASE WHEN rating = 5 THEN 1 ELSE 0 END) * 100.0 / COUNT(*) AS very_good_pct
12
13
14 FROM feedback_rating GROUP BY quarter_number ORDER BY quarter_number;
15
```

Result Grid							
Filter Rows:		Export:		Wrap Cell Content:			
quarter_number	total_feedback	very_bad_pct	bad_pct	okay_pct	good_pct	very_good_pct	
1	310	10.96774	11.29032	19.03226	28.70968	30.00000	
2	262	14.88550	14.12214	20.22901	22.13740	28.62595	
3	229	17.90393	22.70742	21.83406	20.96070	16.59389	
4	199	30.65327	29.14573	20.10050	10.05025	10.05025	

Observations & Insights:

1. Positive feedback collapses from Q1 to Q4. The combined "Good" + "Very Good" share falls from 58.7% in Q1 all the way down to just 20.1% in Q4 a drop of nearly 40 percentage points in a single year. By Q4, less than 1 in 5 customers is satisfied.
2. Negative sentiment nearly triples. "Very Bad" + "Bad" combined goes from 22.3% in Q1 29.0% in Q2 - 40.6% in Q3 - 59.8% in Q4. In Q4, almost 60% of all feedback is negative a critical red flag.
3. Q4 is an alarm state. In Q4, "Very Bad" alone accounts for 30.65% of all feedback making it the single largest feedback category that quarter. This is a complete reversal from Q1 where "Very Good" (30%) was dominant.
4. The "Okay" segment stays relatively stable (19 to 20%) across all quarters, suggesting a fixed neutral customer base. The real shift is the mass migration from positive to negative categories.

5. Total feedback volume is also declining from 310 (Q1), 262 (Q2) ,229 (Q3) ,199 (Q4). Fewer orders in later quarters could mean customers are churning, which compounds the dissatisfaction signal.

Question 6:

What is the trend of the number of orders by quarter?

Solution Query:

```
select quarter_number,count(order_id) as total_count_per_quarter from order_t  
  
group by quarter_number order by quarter_number asc
```

Output:

```
1 • select quarter_number,count(order_id) as total_count_per_quarter from order_t  
2  
3 group by quarter_number order by quarter_number asc  
4
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	quarter_number	total_count_per_quarter		
▶	1	310		
	2	262		
	3	229		
	4	199		

Observations & Insights:

1. Orders are falling every single quarter without exception. The trend is consistently downward 310 , 262 , 229 , 199. There is no recovery or plateau anywhere in the year.
2. Overall decline of 35.8% from Q1 to Q4. New Wheels lost more than a third of its order volume in just one year. This is a significant business contraction that warrants immediate investigation.
3. The quarter-over-quarter (QoQ) drop is remarkably consistent. Each quarter sees roughly a 12to16% decline from the previous one, suggesting this isn't a one-time shock but a structural problem that is compounding over time.
- 4.Strong correlation with satisfaction decline (Q5). The order volume drop mirrors the feedback deterioration seen in Question 5 almost exactly. As satisfaction fell, so did order counts suggesting unhappy customers are churning and not returning for repeat purchases.
5. Q4 with only 199 orders is a critical low. If this trajectory continues into the next year, order volumes could fall below 150 per quarter, threatening revenue sustainability.

Question 7:

Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
WITH CTE AS (  
  
SELECT QUARTER_NUMBER, SUM((VEHICLE_PRICE - DISCOUNT) * QUANTITY) AS  
NET_REVENUE  
  
FROM order_t  
  
GROUP BY QUARTER_NUMBER  
  
)
```

```
SELECT QUARTER_NUMBER, NET_REVENUE, LAG(NET_REVENUE) OVER (ORDER BY
QUARTER_NUMBER) AS PREV_REVENUE, ((NET_REVENUE - LAG(NET_REVENUE) OVER (ORDER
BY QUARTER_NUMBER)) / NULLIF(LAG(NET_REVENUE) OVER (ORDER BY QUARTER_NUMBER),
0)) * 100
```

```
AS QOQ_PERCENT_CHANGE
```

```
FROM CTE ORDER BY QUARTER_NUMBER;
```

Output:

```
1
2 • WITH CTE AS (
3   SELECT QUARTER_NUMBER, SUM((VEHICLE_PRICE - DISCOUNT) * QUANTITY) AS NET_REVENUE
4   FROM order_t
5   GROUP BY QUARTER_NUMBER
6 )
7 SELECT QUARTER_NUMBER, NET_REVENUE, LAG(NET_REVENUE) OVER (ORDER BY QUARTER_NUMBER) AS PREV_RI
8 AS QOQ_PERCENT_CHANGE
9 FROM CTE ORDER BY QUARTER_NUMBER;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	QUARTER_NUMBER	NET_REVENUE	PREV_REVENUE	QOQ_PERCENT_CHANGE
1	1	39637378.16	NULL	NULL
2	2	32913497.44	39637378.16	-16.963485
3	3	29435188.49	32913497.44	-10.568032
4	4	23495814.14	29435188.49	-20.177803

Result 45

Observations & Insights:

1. Net revenue declines every quarter without exception. From 39.6M in Q1 down to 23.5M in Q4 a total fall of 16.1M or 40.7% in a single year. This is a severe and sustained revenue erosion.
2. Q4 sees the steepest QoQ drop at 20.18%. After a relatively moderate Q3 decline of -10.57%, Q4 accelerates sharply downward again. This worsening in Q4 is consistent with the sharp spike in negative feedback seen in Q5 dissatisfied customers likely didn't return.
3. Revenue falls faster than order volume. In Q6, orders declined 35.8% from Q1 to Q4. But revenue declined 40.7% over the same period. This means revenue per order is also shrinking likely due to increasing discounts or a shift toward lower priced vehicle models in later quarters.
4. Q1 correctly shows NULL for both PREV_REVENUE and QOQ_PERCENT_CHANGE since there is no prior quarter.

Question 8:

What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT QUARTER_NUMBER,COUNT(ORDER_ID) AS ORDER_COUNT,SUM((VEHICLE_PRICE-  
DISCOUNT)*Quantity) AS NET_REVENUE
```

```
FROM order_t
```

```
GROUP BY QUARTER_NUMBER
```

```
ORDER BY QUARTER_NUMBER ASC;
```

Output:

```
1 • SELECT QUARTER_NUMBER,COUNT(ORDER_ID) AS ORDER_COUNT,SUM((VEHICLE_PRICE-DISCOUNT)*Quantity) AS NET_REVENUE
2 FROM order_t
3 GROUP BY QUARTER_NUMBER
4 ORDER BY QUARTER_NUMBER ASC;
5
6
```

result Grid			Filter Rows:	Export:	Wrap Cell Content:
QUARTER_NUMBER	ORDER_COUNT	NET_REVENUE			
1	310	39637378.16			
2	262	32913497.44			
3	229	29435188.49			
4	199	23495814.14			

Observations & Insights:

1. Both orders and revenue decline in perfect lockstep every quarter. The two lines in the chart move downward together, confirming a unified business contraction it is not just a pricing issue or just a demand issue, but both simultaneously.
2. Revenue falls faster than orders a compounding problem. Orders dropped 35.8% from Q1 to Q4, but revenue dropped 40.7%. This gap means average revenue per order is also shrinking quarter over quarter, likely driven by heavier discounts, cheaper vehicle mix, or lower quantity orders in later quarters.
3. Revenue per order is declining.
4. The biggest single quarter revenue loss was Q1 and Q2 (6.7M). This is the sharpest absolute dollar drop, likely reflecting the earliest sign of business stress possibly a service or delivery problem that began eroding confidence early in the year.
5. This query is simpler but complementary to Q7. While Q7 used LAG() to compute QoQ percentages, this query gives the raw counts and sums in a single straightforward GROUP BY, making it easy to layer both metrics side by side for business storytelling. Together, Q6, Q7, and Q8 build a complete picture of business decline.

Question 9:

What is the average discount offered for different types of credit cards?

Solution Query:

```
select b.credit_card_type,avg(a.discount) as Avg_discount from order_t a
inner join customer_t b
on a.customer_id=b.customer_id
group by b.credit_card_type
```

Output:

```
1 • select b.credit_card_type,avg(a.discount) as Avg_discount from order_t a
2   inner join customer_t b
3   on a.customer_id=b.customer_id
4   group by b.credit_card_type
5
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
credit_card_type	Avg_discount			
jcb	0.607382			
visa-electron	0.623469			
switch	0.610233			
diners-club-carte-blanche	0.614490			
laser	0.643846			
china-unionpay	0.622174			
diners-club-enroute	0.599792			

Observations & Insights

1. Discounts are remarkably uniform across all card types. The entire range spans only 0.044 from a low of 0.5998 (Diners Enroute) to a high of 0.6438 (Laser). On vehicle purchases worth tens of thousands of dollars, this difference is essentially negligible.
2. Laser card holders receive the highest average discount (0.6438). This is the only card type noticeably above the overall average. It may reflect a coincidental skew in the data (Laser card users happened to buy vehicles with more discounts) rather than a deliberate pricing policy.
3. Diners Enroute receives the lowest discount (0.5998). Again, the difference from the mean is tiny and likely not a strategic decision by New Wheels.
4. There is no meaningful credit card based discount strategy in place. If New Wheels had a loyalty or partnership program tied to specific card types (e.g., higher discounts for premium cards like Visa Electron or UnionPay), we would expect a much larger spread. The near flat distribution suggests discounts are applied uniformly regardless of payment method.
5. . Joining order_t with customer_t on customer_id and grouping by credit_card_type correctly attributes each order's discount to the card type used by that customer.

Question 10:

What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT QUARTER_NUMBER,AVG(DATEDIFF(ship_date, order_date)) AS avg_shipping_time
FROM order_t
GROUP BY QUARTER_NUMBER
ORDER BY QUARTER_NUMBER ASC ;
```

Output:

```
1 • SELECT QUARTER_NUMBER,AVG(DATEDIFF(ship_date, order_date)) AS avg_shipping_time
2 FROM order_t
3 GROUP BY QUARTER_NUMBER
4 ORDER BY QUARTER_NUMBER ASC ;
```

	QUARTER_NUMBER	avg_shipping_time
▶	1	57.1677
	2	71.1107
	3	117.7555
	4	174.0955

Observations & Insights

1. Shipping time tripled over the year the most alarming finding so far. Average shipping time exploded from 57 days in Q1 to 174 days in Q4, a 204.5% increase. By Q4, customers are waiting nearly 6 months for delivery after placing an order.
2. The acceleration is getting worse each quarter. The jump from Q2 and Q3 alone was +65.6% (from 71 to 118 days). This is not a stable situation it is a runaway logistics crisis.
3. As shipping time goes up, satisfaction goes down, negative feedback spikes, and orders/revenue follow. Shipping delays are the most likely primary driver of all business deterioration seen in Q4 and Q8.
4. Q4 at 174 days (25 weeks) is operationally unacceptable for vehicle delivery. Customers waiting 6 months have every reason to cancel, seek competitors, and leave negative reviews.

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
125,482,878	1,000	1,000	3.08 / 5
Last Quarter Revenue	Last Quarter Orders	Average Days to Ship	% Good Feedback
23,495,814	199	105 days	39.1%

Observations & Insights

- 1. Average shipping time exploded from 57 days in Q1 to 174 days in Q4, a 204.5% increase. By Q4, customers are waiting nearly 6 months for delivery after placing an order.
- 2. The acceleration is getting worse each quarter. The jump from Q2 and Q3 alone was +65.6% (from 71 to 118 days). This is not a stable situation it is a runaway logistics crisis.
- 4. Q4 at 174 days (25 weeks) is operationally unacceptable for vehicle delivery. Customers waiting 6 months have every reason to cancel, seek competitors, and leave negative reviews.

Business Recommendations

1. Fix the logistics crisis urgently. Shipping time tripled from 57 days (Q1) to 174 days (Q4) the root cause of satisfaction collapse. Target delivery under 60 days through shipper audits and supply chain restructuring.
2. Investigate Q3 and Q4 operational breakdown. All key metrics deteriorate sharply from Q3 onward. A root cause analysis of staffing, inventory, and shipper capacity is critical to prevent recurrence.
3. Leverage top states for targeted retention. Texas, California, and Florida drive the most orders. Priority delivery guarantees and loyalty campaigns in these states can stabilize the customer base.
4. Build maker partnerships with Chevrolet and Ford. These two brands dominate national orders. Securing priority inventory and comarketing deals will protect order volumes, especially in weaker quarters.
5. Introduce differentiated discount tiers by card type. Current discounts are near identical across all credit card types (0.60). Tiered card linked offers can drive repeat purchases and improve revenue per order.
6. Target an average rating recovery of ≥ 3.5 . Q4 satisfaction hit a critical low of 2.40. Proactive delivery communication, after sales support, and faster shipping will be key levers to restore customer trust.